

Geometry

Unit 3

Lesson 6 Homework

Name **Answer Key** \_\_\_\_\_

Date \_\_\_\_\_

Period \_\_\_\_\_

$\overleftrightarrow{MN} \parallel \overleftrightarrow{PH}$ , where  $\overleftrightarrow{RV}$  is a transversal with  
 $m\angle RSM = (12x - 7)^\circ$ , and  $m\angle PTV = (4x - 5)^\circ$  as shown.

1. Determine the value of  $x$ .

$x = 12$

2. What is  $m\angle RSM$ ?

$137^\circ$

3. What is  $m\angle PTV$ ?

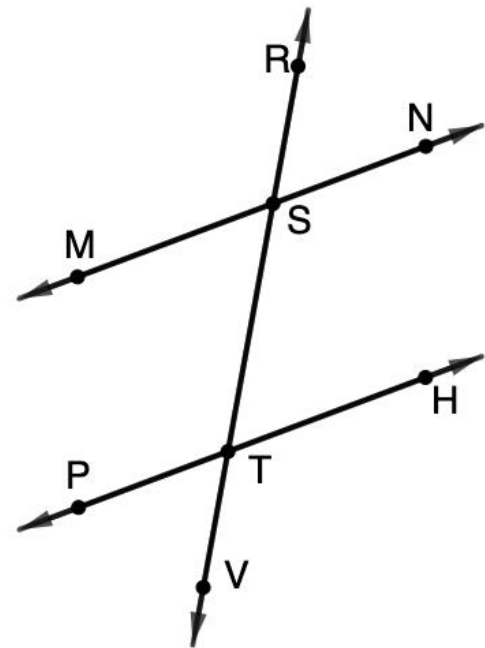
$43^\circ$

4. List all the angles congruent to  $\angle RSM$ .

$\angle RSM \cong \angle NST \cong \angle HTV \cong \angle PTS$

5. List all the angles congruent to  $\angle PTV$ .

$\angle PTV \cong \angle STH \cong \angle RSN \cong \angle MST$



$\overleftrightarrow{MN} \parallel \overleftrightarrow{PH}$ , where  $\overleftrightarrow{RV}$  is a transversal with  $m\angle RSN = (3x - 17)^\circ$  and  $m\angle STP = (7x - 3)^\circ$  as shown.

6. Determine the value of  $x$ .

$x = 20$

7. What is  $m\angle RSN$ ?

$43^\circ$

8. What is  $m\angle STP$ ?

$137^\circ$

9. What is  $m\angle RSM$ ?

$137^\circ$

10. What is  $m\angle HTS$ ?

$43^\circ$

